

CSE4227 Digital Image Processing

Lecture 03 – Spatial and Gray Level Resolutions

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Today's Contents

□ Image Digitization

- Spatial Resolution
- Gray Level Resolution
- Representation of Digital Image
- Effect of Spatial and Gray Level Resolutions
- Image Scaling (Zooming & Shrinking)

• Chapter 2 from R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008 [**Section 2.4, except 2.2.4**]

Human Vision VS Computer Vision



What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

A Simple Image Formation Model

$$f(x, y) = i(x, y) \cdot r(x, y)$$

$f(x, y)$: intensity at the point (x, y)

$i(x, y)$: illumination at the point (x, y)

(the amount of source illumination incident on the scene)

$r(x, y)$: reflectance/transmissivity at the point (x, y)

(the amount of illumination reflected/transmitted by the object)

where $0 < i(x, y) < \infty$ and $0 < r(x, y) < 1$

Representation of Digital Image

Formally,

- Sampling:

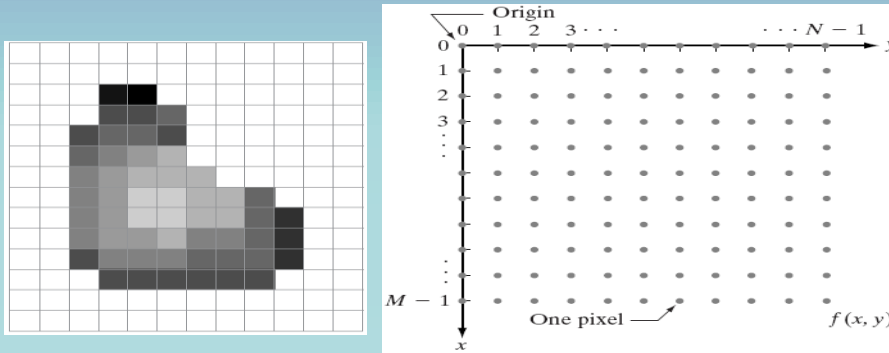
- Dividing (x, y) plane into grid with coordinate (z_i, z_j) , where,

$$z_i, z_j \in Z$$

- Quantization:

- Assign gray level value from Z to $f(z_i, z_j)$

Spatial Resolution



❑ **Spatial resolution:** $M \times N$

❑ Number of Rows (M) and Columns (N) can be any integer

❑ Bit size of an image: $b = M \times N \times k$

❑ For Square image, $b = N^2 k$

An image that is 2048 pixels in width and 1536 pixels in height has a total of $2048 \times 1536 = 3,145,728$ pixels or 3.1 megapixels.

Here, spatial resolution is 3.1-megapixel.

One could refer to it as 2048 by 1536 or a 3.1-megapixel image.

Intensity (Gray) Level Resolution

Intensity level resolution refers to the number of intensity levels used to represent the image

❑ **Intensity (Gray) level resolution:** L

❑ Number of gray levels, L , is usually power of 2:

❑ k = Bit depth $L = 2^k$

❑ Gray level range = $[0 - L-1]$
= $[0 - 255]$

Where $k = 8$

$$L = 2^8$$

$$L = 256.$$

Intensity (Gray) Level Resolution

- ❑ The more intensity levels used, the finer the level of detail discernable in an image
- ❑ Intensity level resolution is usually given in terms of the number of bits used to store each intensity level

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

Number of storage of bits:

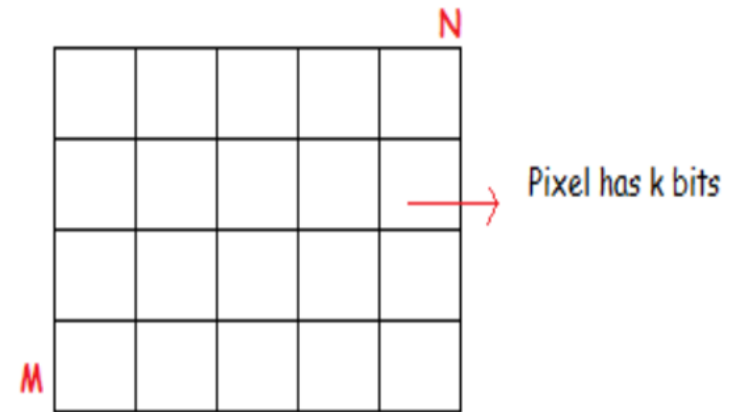
$N * M$: the no. of pixels in all the image.

K : no. of bits in each pixel

L : grayscale levels the pixel can represent

$$L = 2^K$$

all bits in image = $N * M * k$



NO of pixels = $N * M$

NO of bits = $N * M * k$

Image Sizes in bits for Different Spatial and Gray Level Resolutions

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

❑ The calculation is for square images

❑ N = number of row/column

❑ L = gray level resolution

❑ K = bit required to represent gray levels

Answer the questions:

Q1: Suppose a pixel has 1 bit, how many gray levels can it represent?

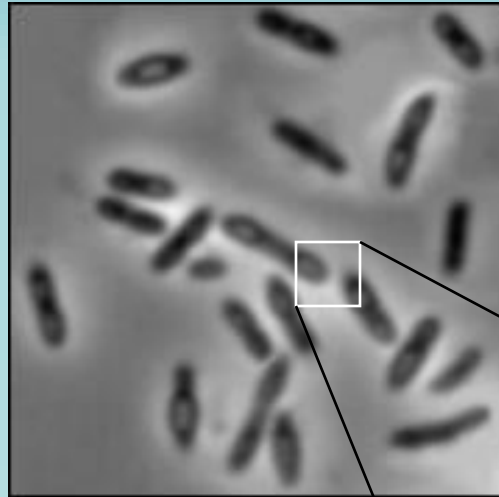
Q2: Suppose a pixel has 2 bit, how many gray levels can it represent?

Q3: Suppose a pixel has p bit, how many gray levels can it represent?

Q4: if we want to represent 256 intensities of grayscale, how many bits do we need?

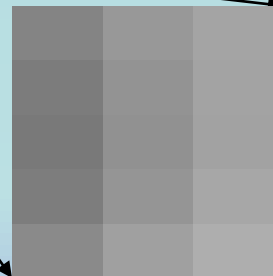
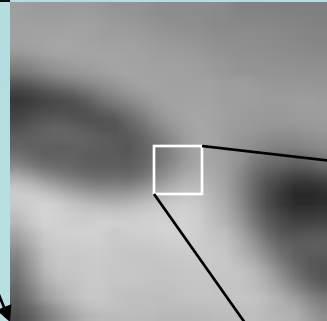
Q5: What is the **Gray Level Resolution** and **Spatial Resolution** of a 3 bit 5x5 image? What will be the image size in bits? How many gray levels can be there in the image? What is bit depth?

Digital Image Types : Intensity Image

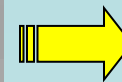


Intensity image or monochrome image

each pixel corresponds to light intensity
normally represented in gray scale (gray level).



Gray scale
values



10	10	16	28
9	6	26	37
15	25	13	22
32	15	87	39



components

components


$$\begin{bmatrix} 10 & 65 & 70 & 56 & 43 \\ 9 & 6 & 99 & 26 & 70 & 37 & 56 & 78 \\ 15 & 32 & 9 & 34 & 96 & 67 & 67 \\ 32 & 21 & 60 & 13 & 90 & 22 & 96 & 67 \\ & 54 & 15 & 85 & 87 & 85 & 39 & 43 & 39 & 92 \\ & & & 32 & 65 & 87 & 99 \end{bmatrix}$$

Image Types : Binary

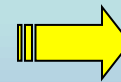
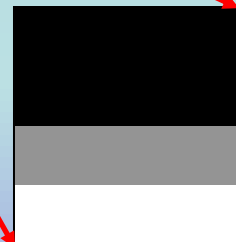
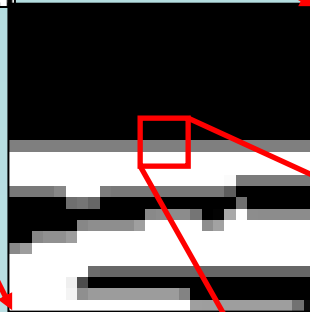
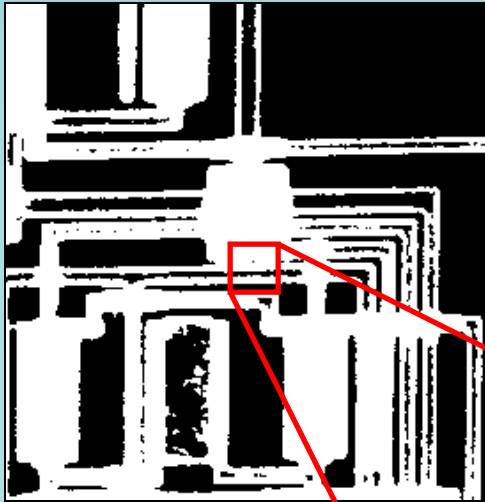
Image

Binary image or black and white image

Each pixel contains one bit :

1 represent white

0 represents black



Binary

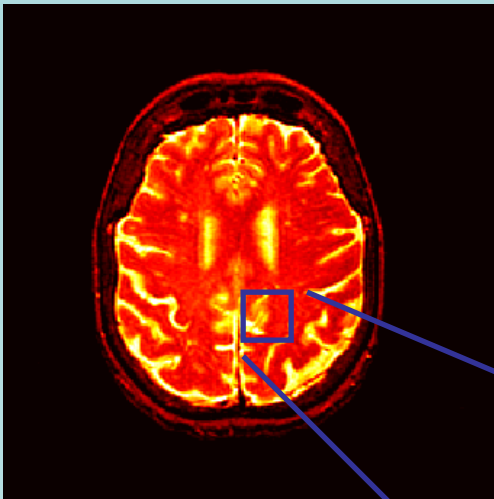
data

0	0	0	0
0	0	0	0
1	1	1	1
1	1	1	1

Image Types : Index Image

Index image

Each pixel contains index number pointing to a color in a color table



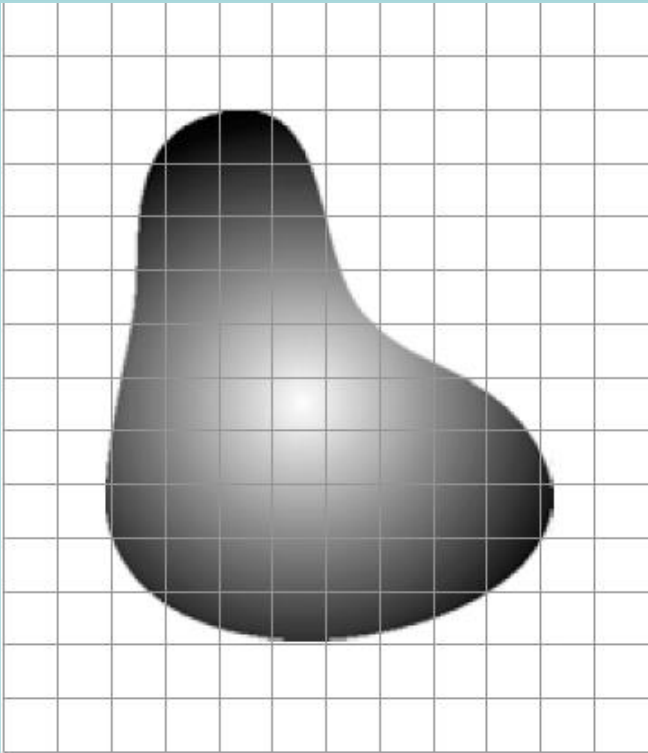
$\begin{bmatrix} 1 & 4 & 9 \\ 6 & 4 & 7 \\ 6 & 5 & 2 \end{bmatrix}$

Index
value

Color Table

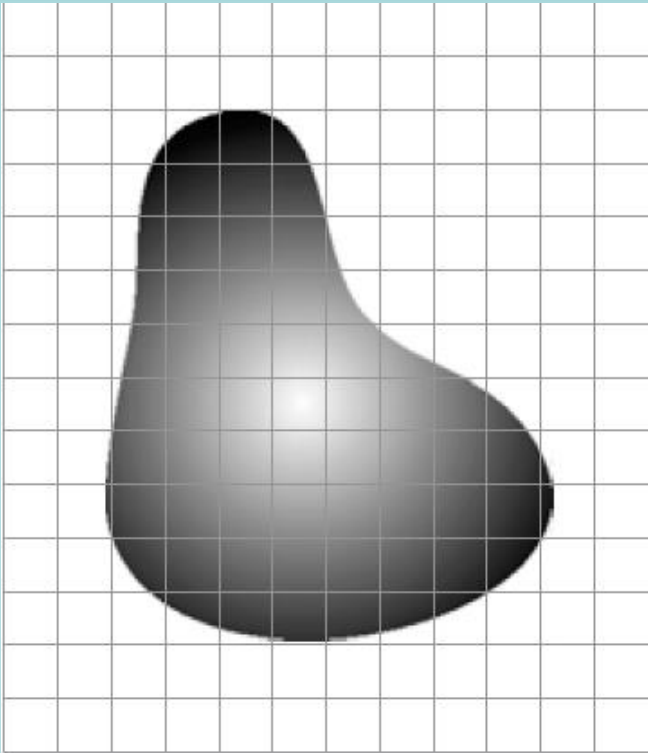
Index No.	Red component	Green component	Blue component
1	0.1	0.5	0.3
2	1.0	0.0	0.0
3	0.0	1.0	0.0
4	0.5	0.5	0.5
5	0.2	0.8	0.9
...	...	...	...

Effect of Spatial and Gray Level Resolutions



- ❑ 14X12 resolution means samples from **14X12**
- ❑ What happens, if we sample from **8X8** locations?

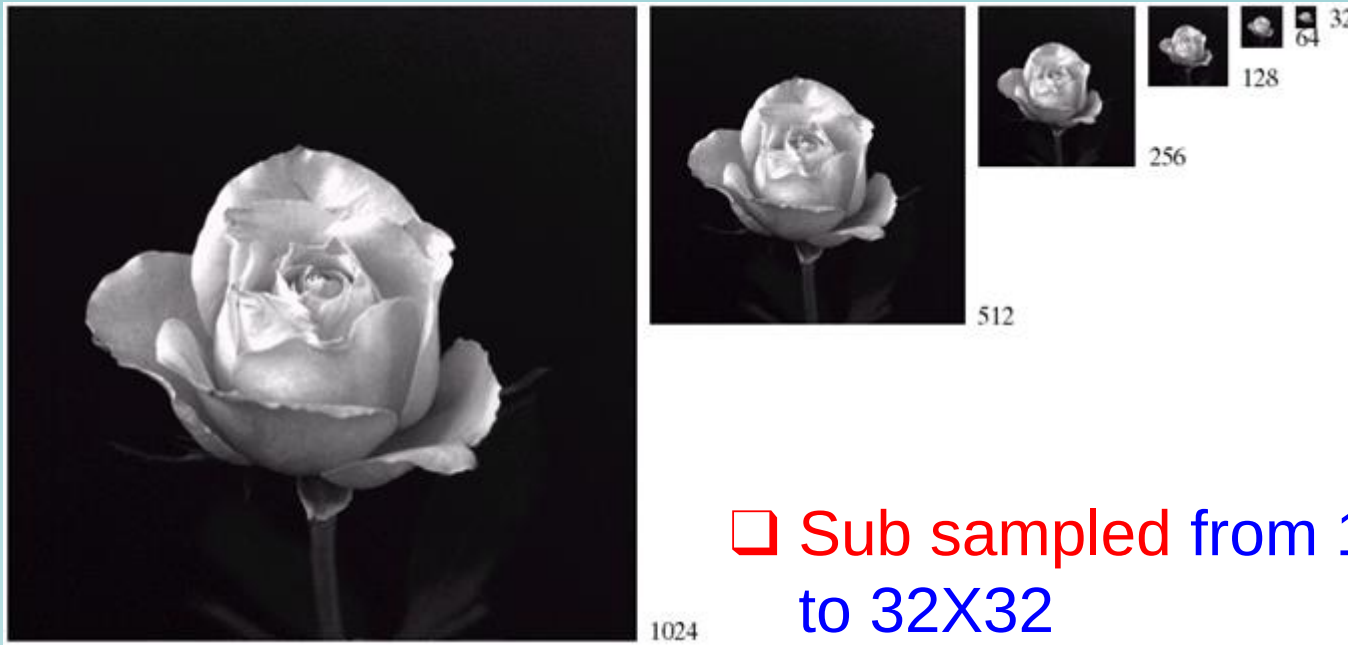
Effect of Spatial and Gray Level Resolutions



- ❑ 14X12 resolution means samples from **14X12**
- ❑ What happens, if we sample from **8X8** locations?

- ❑ Larger cell size
- ❑ Lower spatial resolution
- ❑ Lower features spatial accuracy
- ❑ Lower file size, faster display and faster processing time

Effect of Spatial Resolutions



❑ Sub sampled from 1024X1024 up to 32X32

❑ unchanged gray level (intensity)

❑ **Delete** alternate row and column while sub sampling

Effect of Spatial Resolutions

Resized to 1024 x 1024

From → 1024X1024 512X512 256X256



Using column
duplication process



From → 128X128 64X64 32X32

Can we increase spatial resolution ?

From $\longrightarrow$ 128X128 64X64 32X32

Using nearest
neighbor



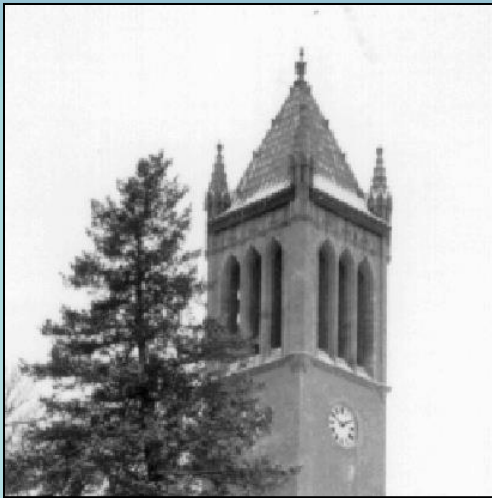
Using bilinear
interpolation



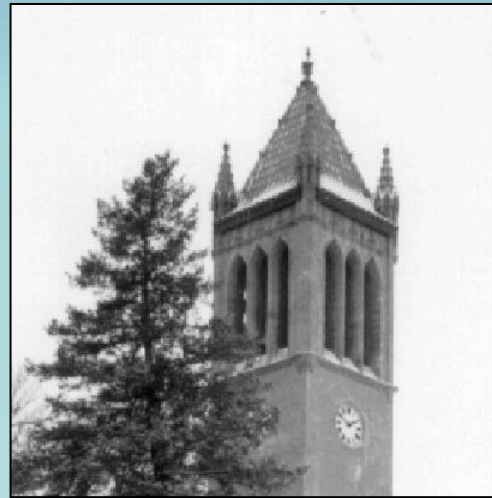
From $\longrightarrow$ 128X128 64X64 32X32

Down sampling is an irreversible process.

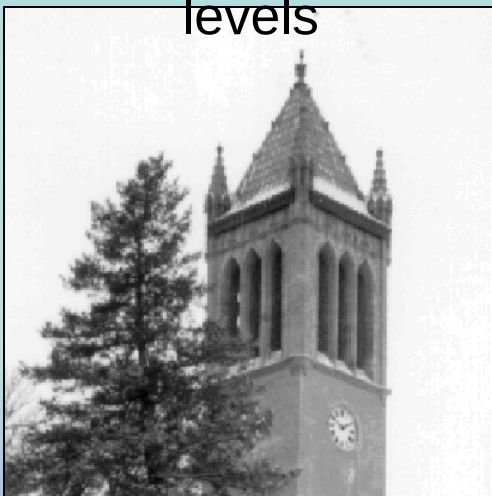
Effect of Gray level Resolutions



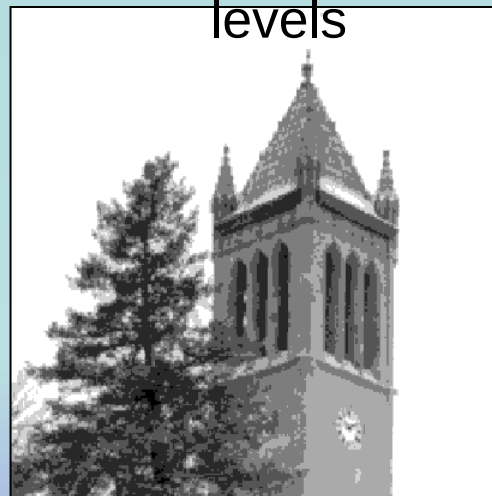
256
levels



128
levels



64
levels

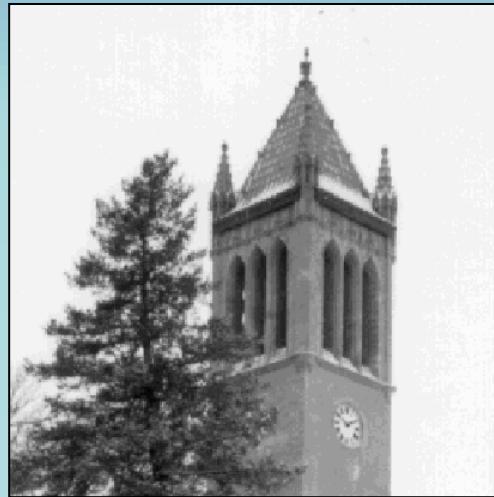


32
levels

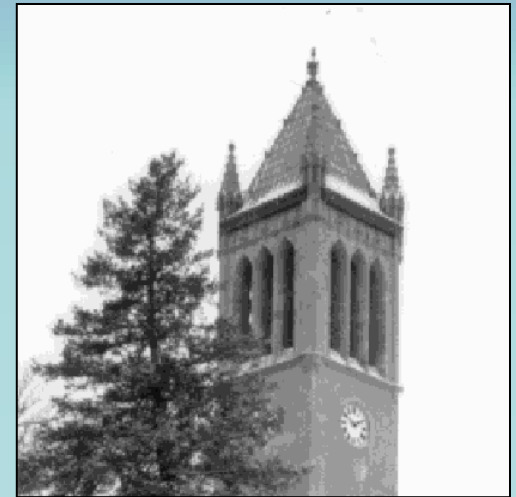
- unchanged spatial resolution
- Gray level changed from 256 to 32

Effect of Gray level Resolutions

□ Gray level
changed
from 16 to 2

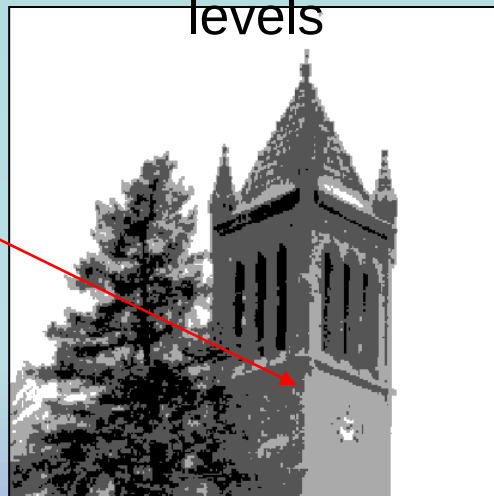


16
levels

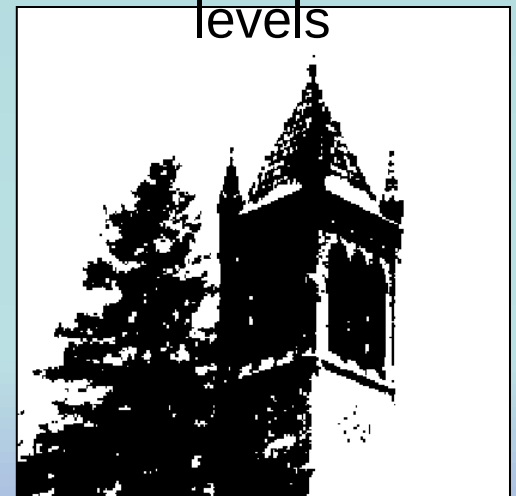


8
levels

In this
image,
it is easy to
see
false contour.

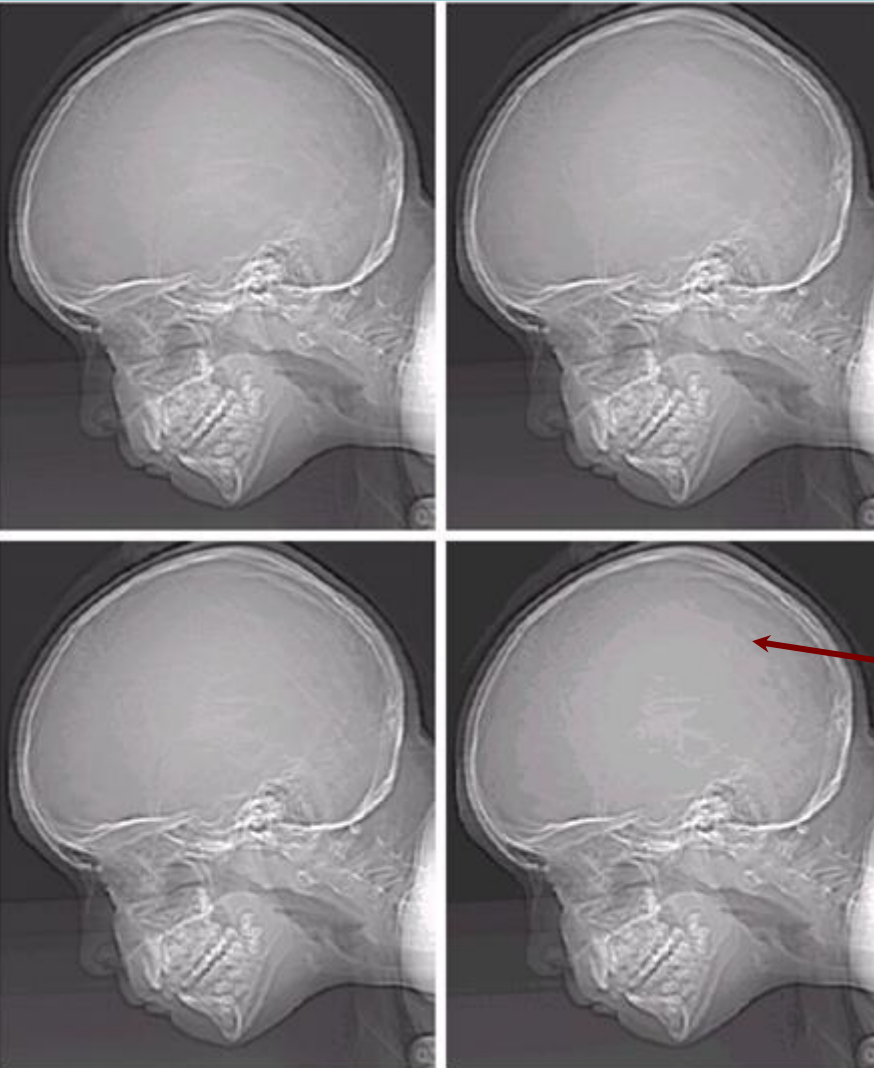


4
levels



2
levels

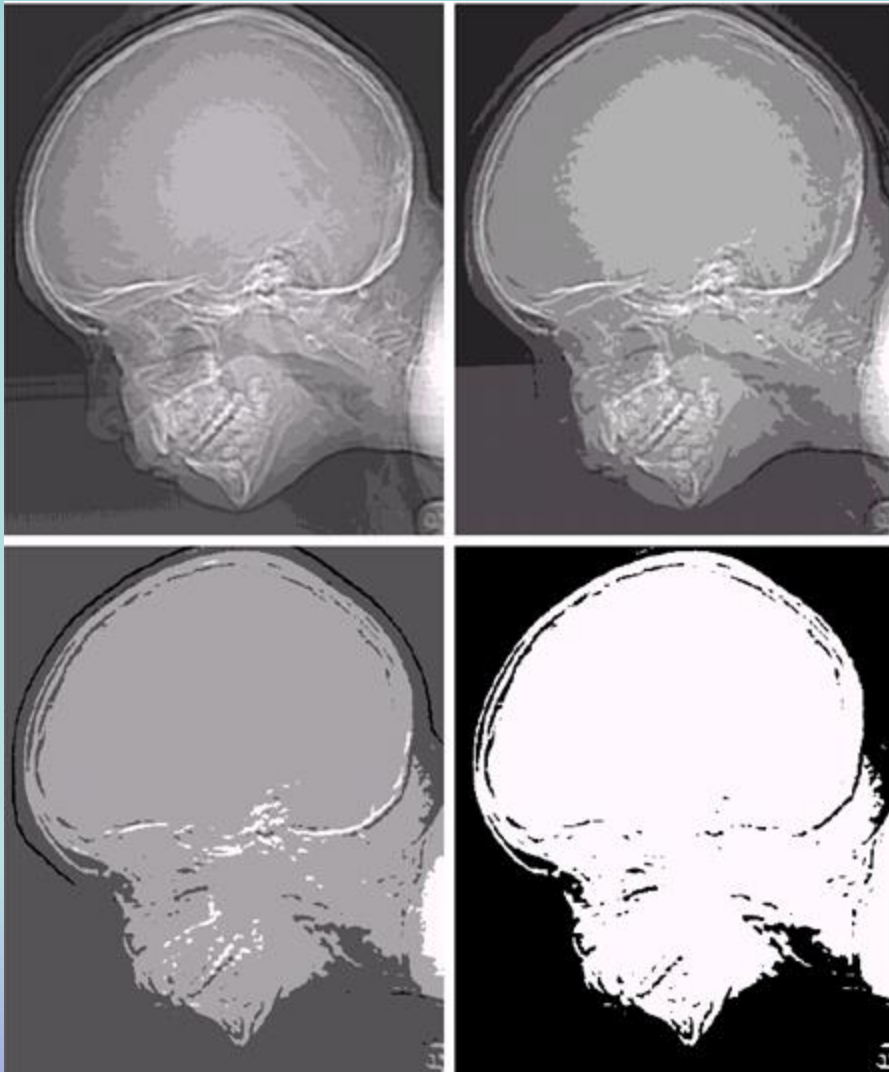
Effect of Gray level Resolutions



- unchanged spatial resolution
- Gray level changed from 256 to 32

Ridge-like structure in the smooth area

Effect of Spatial and Gray Level Resolutions



- Gray level changed from 16 to 2
- Ridge-like structure is more prominent
- Reason: insufficient number of gray levels used

Resolution: How Much Is Enough?

□ The big question with resolution is always *how much is enough?*

- This all depends on what is in the image and what you would like to do with it
- Key questions include
 - Does the image look aesthetically pleasing?
 - Can you see what you need to see within the image?

Resolution: How Much Is Enough? (cont...)



- ❑ The picture on the right is fine for counting the number of cars, but not for reading the number plate

How to select the suitable size and pixel depth of images

The word “suitable” is subjective: depending on “subject”.



Low detail image

Lena image



Medium detail image

Cameraman image



High detail image

To satisfy human mind

1. For images of the same size, the low detail image may need more pixel depth.
2. As an image size increase, fewer gray levels may be needed.

Image Scaling (Zooming and Shrinking)

- **Zooming (up scaling, resizing upward) requires two steps**
 - ❑ The Creation of new pixel locations
 - ❑ Assignment of gray levels to new pixel locations

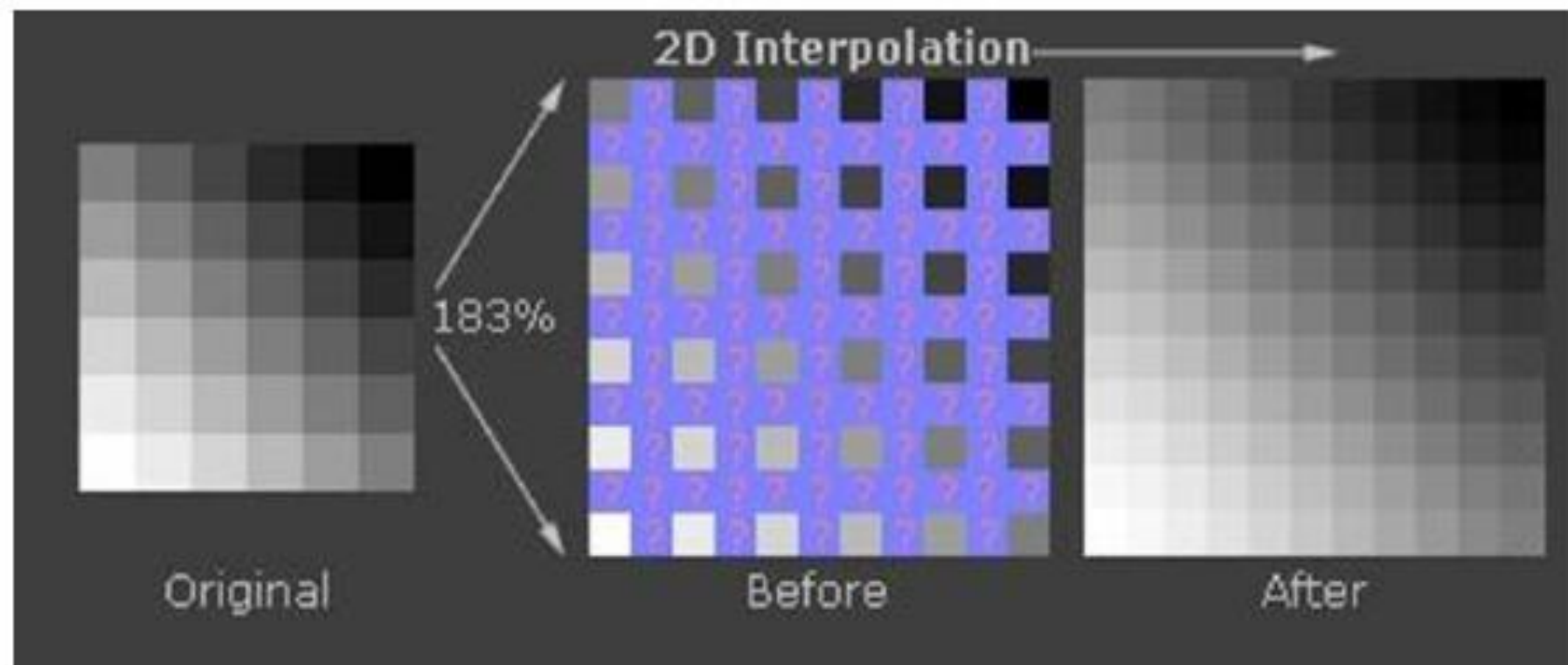


Image Scaling (Zooming and Shrinking)

- Zooming (up scaling, resizing upward) can be achieved by the following techniques::

- ❑ Nearest neighbor Interpolation

- ❑ Pixel Replication

- ❑ Bilinear Interpolation

- ❑ Bicubic Interpolation

Image Scaling (Zooming and Shrinking)

▪ Pixel Replication

- ❑ Pixel replication is applicable when we want to increase the size of an image an integer number of times.
- ❑ For example to double the size of an image we can duplicate each column, this doubles the size of image in horizontal direction. Then we duplicate each row of the enlarged image to double the size in the vertical direction
- ❑ The same procedure can be applied to enlarge the image by any integer number of times (triple, quadruple and so on)
- ❑ The gray level assignment of each pixel is predetermined by the fact that new locations are exact duplicate of old locations

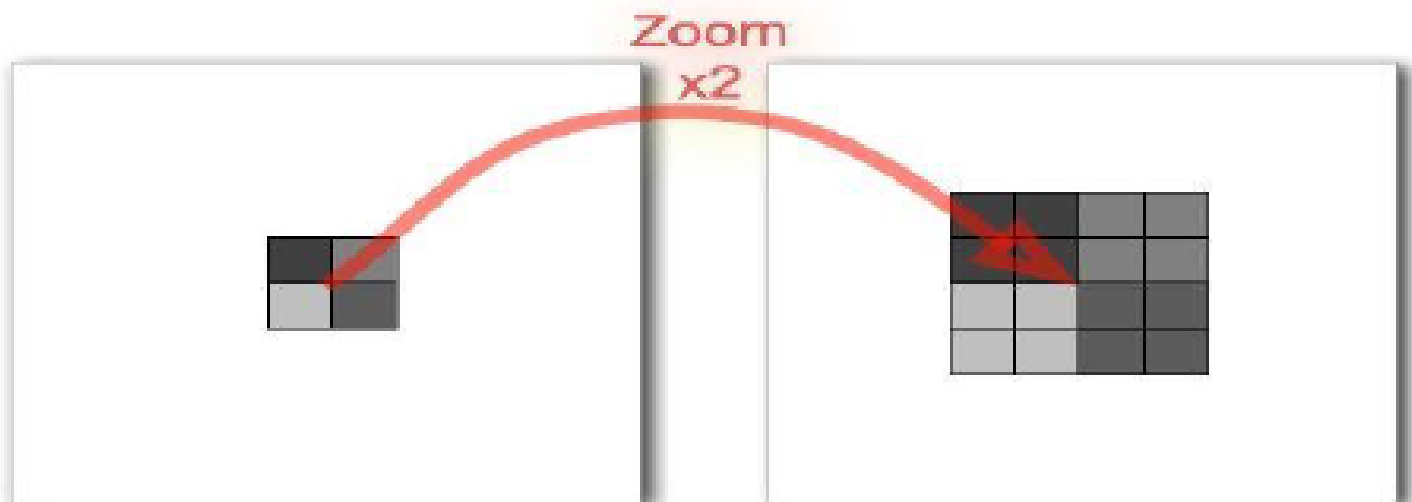
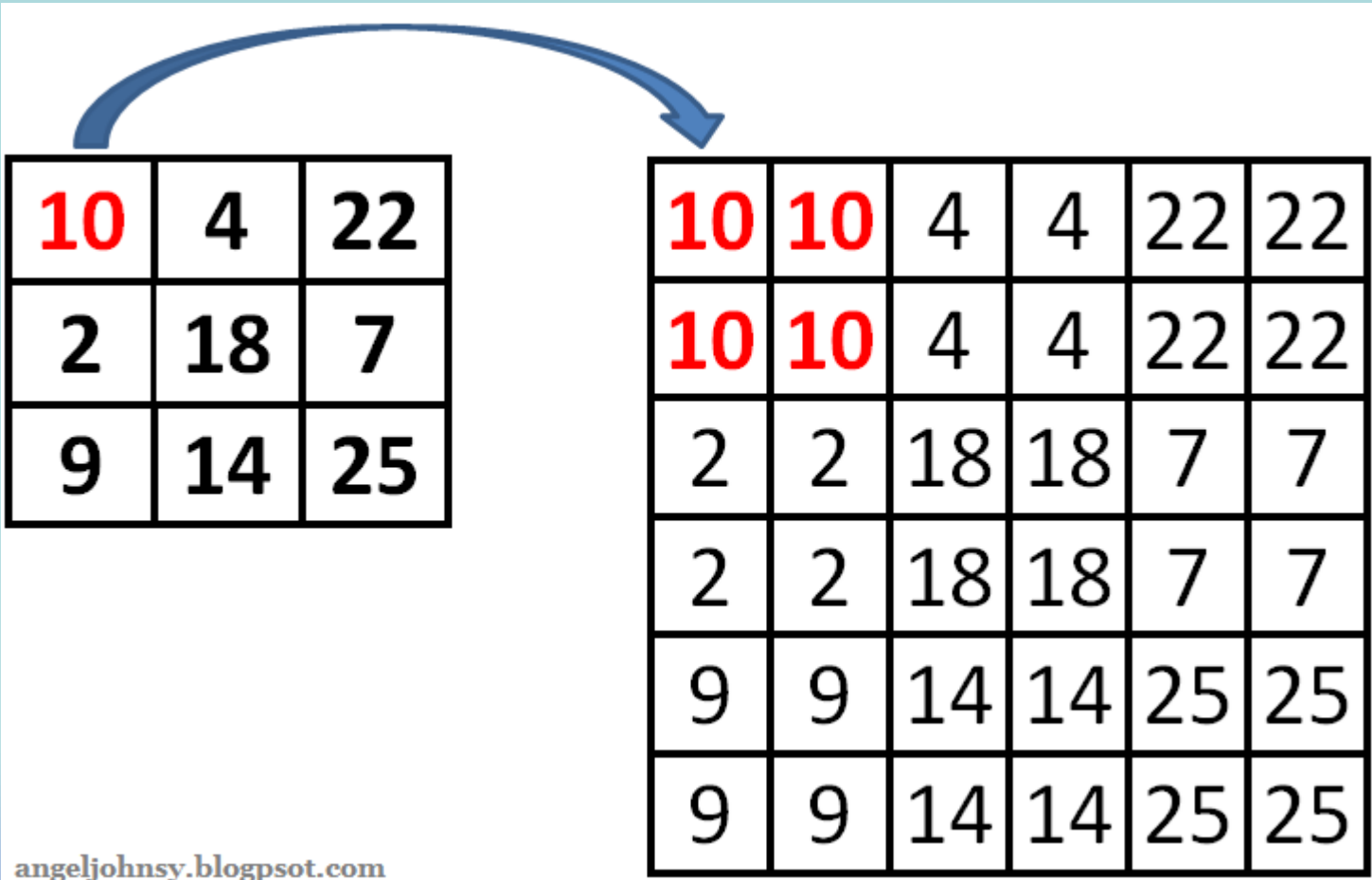


Image Scaling (Zooming and Shrinking)

▪ Nearest neighbor Interpolation

- ❑ Suppose that we have an image of size 500 x 500 and we want to enlarge it to 1.5 times 750 x 750 pixels.
- ❑ For any zooming approach we have to create an imaginary grid of the size which is required over the original image. In that case we will have an imaginary grid of 750 x 750 over an original image.
- ❑ Obviously the spacing in the grid would be less than one pixel because we fitting it over a smaller image. In order to perform gray level assignment for any point in the overlay, we look for the closest pixel in the original image and assign its gray level to new pixel in the grid.
- ❑ When finished with all points in the grid, we can simply expand it to the originally specified size to obtain the zoomed image. **This method of gray level assignment is called nearest neighbor interpolation**

3x3 matrix is interpolated to 6x6 matrix



Steps to be performed:

1. Consider a matrix $A = \begin{bmatrix} 10 & 4 & 22 \\ 2 & 18 & 7 \\ 9 & 14 & 25 \end{bmatrix}$ where number of rows = 3 and number of columns = 3

2. Define the new size for the matrix , number of rows = 6 and number of columns = 6

3. Find the Ratio of the new size and old size,

$$Ratio_{Row} = \frac{3}{6} \text{ and } Ratio_{Column} = \frac{3}{6}$$

4. Normalize the row-wise pixel positions and column wise pixel positions based on the new size

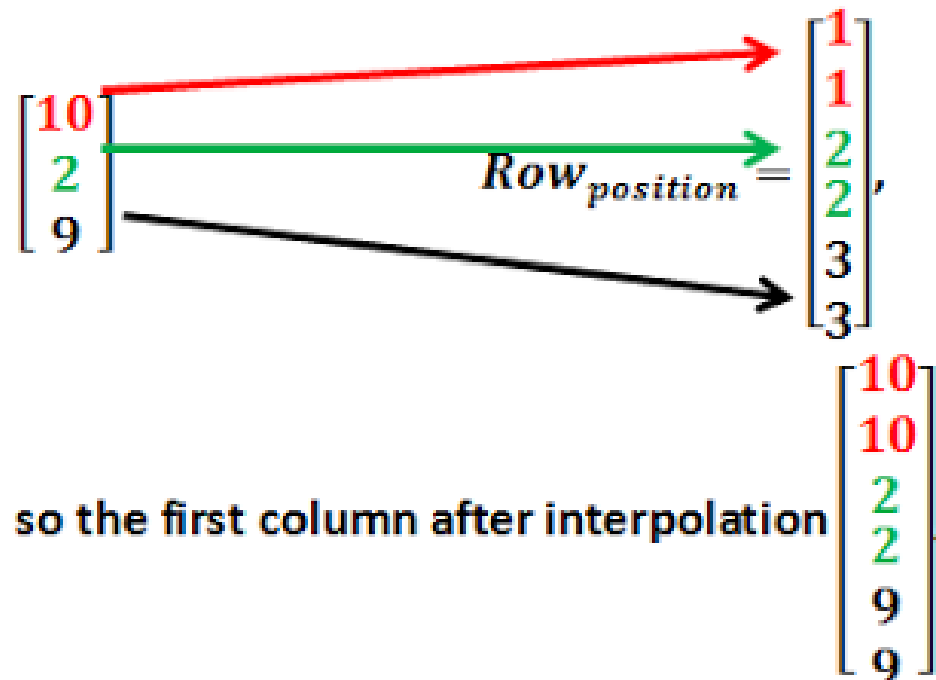
$$Row_{Positions} = \frac{[1 \ 2 \ 3 \ 4 \ 5 \ 6]}{Ratio_{Row}} = [0.5 \ 1 \ 1.5 \ 2 \ 2.5 \ 3]$$

5. By using ceil function that gives the least integer greater than or equal to

the value, the $Row_{position} = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \\ 3 \\ 3 \end{bmatrix}$ and

$Column_{position} = [1 \quad 1 \quad 2 \quad 2 \quad 3 \quad 3]$

6. Perform row-wise interpolation on the columns of the matrix,
For instance, consider the first column of the matrix,



7. Similarly, perform the row-wise interpolation on all the columns of the matrix,

$$\begin{bmatrix} 10 & 4 & 22 \\ 10 & 4 & 22 \\ 2 & 18 & 7 \\ 2 & 18 & 7 \\ 9 & 14 & 25 \\ 9 & 14 & 25 \end{bmatrix}$$

8. Perform column wise interpolation,

Take the first row from the row-wise interpolated matrix $A, [10 \ 4 \ 22]$ and based on the `column_position`, the values are interpolated. The column position 1 indicates the 1st element i.e. 10, 2 indicates the 2nd element (i.e) 4 so the interpolated first row is $[10 \ 10 \ 4 \ 4 \ 22 \ 22]$

First row = $[10 \ 4 \ 22]$, *Column_{position}* = $[1 \ 1 \ 2 \ 2 \ 3 \ 3]$



Interpolated first row is $[10 \ 10 \ 4 \ 4 \ 22 \ 22]$

9. Similarly, perform this column wise interpolation on all the row wise interpolated rows,

10. The interpolated 6x6 matrix is

$$\begin{bmatrix} 10 & 10 & 4 & 4 & 22 & 22 \\ 10 & 10 & 4 & 4 & 22 & 22 \\ 2 & 2 & 18 & 18 & 7 & 7 \\ 2 & 2 & 18 & 18 & 7 & 7 \\ 9 & 9 & 14 & 14 & 25 & 25 \\ 9 & 9 & 14 & 14 & 25 & 25 \end{bmatrix}$$

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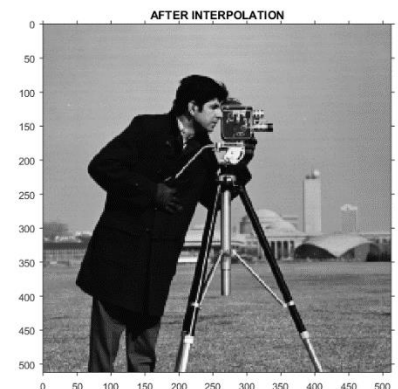
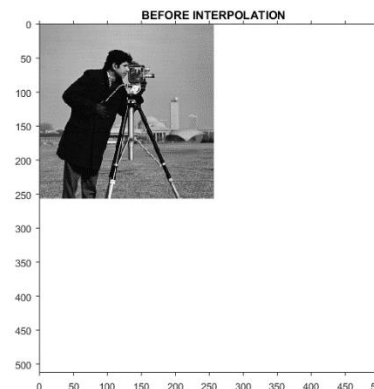
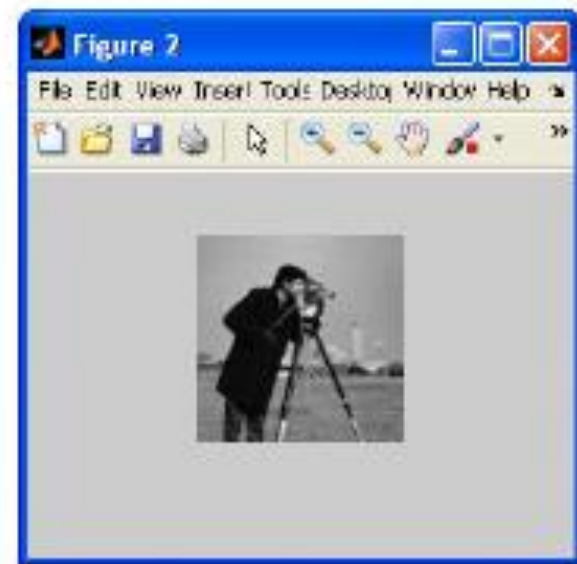


Image Scaling (Zooming and Shrinking)

- **Shrinking (Down scaling, resizing downward)**

- ❑ Image shrinking is done in the similar manner as zooming with one difference as now the process of pixel replication is row column deletion. Now we can delete every second column and row for shrinking



Class Work:

- Interpolate a 2x2 matrix [1, 4, 3, 7] to 4x4
- Interpolate a 3x3 matrix [1,4,3,7,2,3,5,6,0] to 9x9
- Interpolate a 2x2 matrix [1, 2, 3, 4] to 8x8
- Interpolate a 3x3 matrix [1,4,3, 7, 5, 6] to 6x6



Home Work

- **How Zooming and Shrinking Images by Pixel Replication?**
 - **Explain**
 - **Write program**

Any question



Image Scaling (Zooming and Shrinking)

▪ Bilinear Interpolation

- ❑ Bilinear interpolation considers the closest 2x2 neighborhood of known pixel values surrounding the unknown pixel.
- ❑ It then takes a weighted average of these 4 pixels to arrive at its final interpolated value. This results in much smoother looking images than nearest neighbor.
- ❑ The diagram below is for a case when all known pixel distances are equal, so the interpolated value is simply their sum divided by four.
- ❑ In case the distance varies then The closer pixels are given more weightage in the calculation

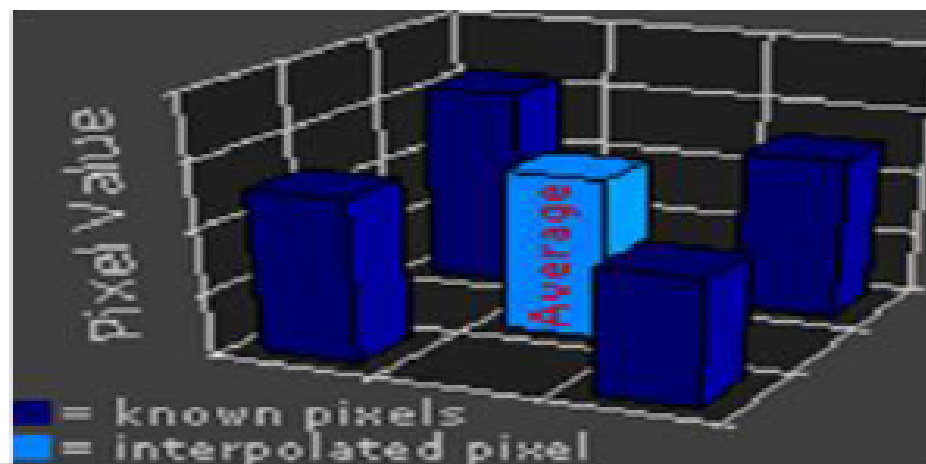


Image Scaling (Zooming and Shrinking)

- The key idea is to perform linear interpolation first in one direction, and then again in the other direction.
- Suppose that we want to find the value of the unknown function f at the point $P = (x, y)$.
- It is assumed that we know the value of f at the four points $Q_{11} = (x_1, y_1)$, $Q_{12} = (x_1, y_2)$, $Q_{21} = (x_2, y_1)$, and $Q_{22} = (x_2, y_2)$.

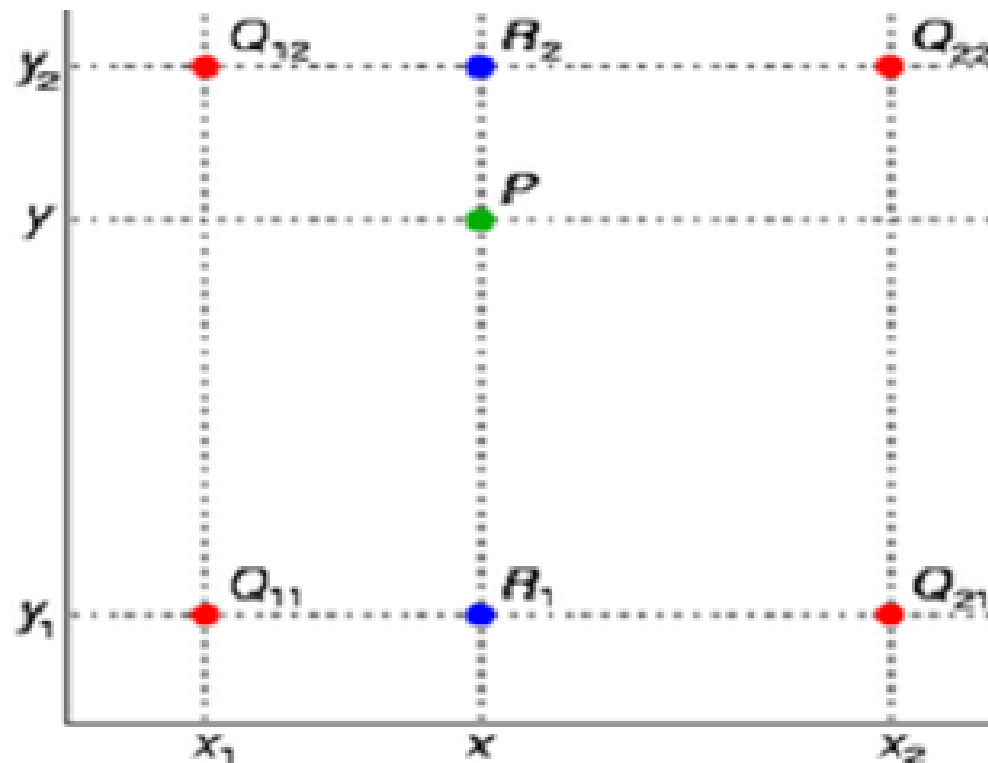


Image Scaling (Zooming and Shrinking)

We first do linear interpolation in the x -direction. This yields

$$f(R_1) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21})$$

where $R_1 = (x, y_1)$,

$$f(R_2) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22})$$

where $R_2 = (x, y_2)$.

$$f(P) \approx \frac{y_2 - y}{y_2 - y_1} f(R_1) + \frac{y - y_1}{y_2 - y_1} f(R_2).$$

We proceed by interpolating in the y -direction.

Image Scaling (Zooming and Shrinking)

This gives us the desired estimate of $f(x, y)$.

$$\begin{aligned} f(x, y) \approx & \frac{f(Q_{11})}{(x_2 - x_1)(y_2 - y_1)}(x_2 - x)(y_2 - y) \\ & + \frac{f(Q_{21})}{(x_2 - x_1)(y_2 - y_1)}(x - x_1)(y_2 - y) \\ & + \frac{f(Q_{12})}{(x_2 - x_1)(y_2 - y_1)}(x_2 - x)(y - y_1) \\ & + \frac{f(Q_{22})}{(x_2 - x_1)(y_2 - y_1)}(x - x_1)(y - y_1). \end{aligned}$$

<https://theailearner.com/2018/12/29/image-processing-bicubic-interpolation//>